Viscar Industrial Capacity Limited
Enhancing Performance & Productivity

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ELECTRICAL SYSTEMS DESIGN TRAINING

COURSE CONTENT

MAIN CONTENT CATEGORY:

1. **Electrical Systems (Definitions & Types)**

- Definition of Electrical Systems
- Definition of Electrical Systems Design
- Types of Electrical Systems

2. **OEMs & Electrical Systems Switchgears/Devices/Parts/Components/Equipment**

OEMs (ORIGINAL EQUIPMENT MANUFACTURERS):

- Allen-Bradley
- ABB
- Chint
- Schneider Electric
- Eaton
- Siemens
- Omron
- Tense
- Minilec
- Salzer
- Connectwell
- Phoenix Contact
- Telemecanique Controls
- Selec
- Pepperl+Fuchs
- GIC ETC.

SWITCHGEARS/EQUIPMENT/DEVICES/PARTS/COMPONENTS

- Protection Devices (Circuit Breakers, Motor Overload Switches, Safety Fuses Etc.)
- Relays, Contactors & Timers

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- Signal Devices
- Sensor, Switch & Pushbuttons
- Measuring Instruments
- Converters (VFDs, Soft starters
- Terminals
- Inductors
- Capacitors
- Voltage Source & Generator
- Transformers
- Cables & Busbars

3**. System Design Information/Technical Specification/Design Parameters**

- Standard
- Current Protection
- Voltage Protection
- Mechanical Information
- Grid Connection Information
- Earthing Systems
- Location Information
- Protection of Persons
- Wiring Color Code
- Wire Sizes

4**. Electrical Systems Design Standards, Codes & Regulations**

Key Regulatory Documents Codes & Standards

- Local Standards
- International Standards
- KEGC (Kenya Electricity Grid Codes)
- EPRA (Energy & Petroleum Regulatory Authority) Regulations
- KEBS (Kenya Bureau of Standards)
- IEC (International Electromechanical Commission)
- BS (British Standards)

5. **Electrical Systems Design Platforms (ECAD Software Tools & Design Strength Properties)**

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EPLAN Electric P8 & EPLAN Pro Panel ECAD SOFTWARE TOOLS

- Schematics - SLDs & MLD
- Cross References
- Cables & Terminals
- Macros
- Device Concept
- Parts Management/Bill of Material
- Navigators
- Mounting Panel
- Forms & Plotframes
- Saving & Sending projects
- Export in PDF, DWG Format
- PLC
- Creating Graphical Report
- Message Management/Checking Schematics

6. **Best Practice - Safe & Efficient Electrical System Design**

- Electrical Systems Symbols & Schematic Interpretation
- Switchgear Selection Criterion & Sizing
- Cable & Busbars Sizing
- Earthling Systems
- Cable Numbers
- SLD & MLD Development
- Layout Development (2D & 3D)
- Labels & Tags
- BoM & Quotation Estimations
- Design Documentation - For Approval/For Manufacture/As-Built
- Electrical System Project Management (RFIs, RFAs etc.)
- Safety Compliance
- Load Balancing
- Scalability
- Energy Optimizations

7**. Electrical Systems Standard Design Procedure**

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- Project Definition
- Site Assessment
- Regulatory Compliance
- Load Analysis
- System Architecture
- Equipment Selection
- Schematic Diagrams
- Cable & Conduit Routing
- Grounding & Bonding
- Protection & Coordination
- Control System Design
- Emergency & Backup Systems
- Documentation
- Review & Validation
- Implementation & Construction
- Testing & Commissioning
- As-Built Documentation
- Training & Handover

8. **Industrial-Grade Systems Design**

- Design of Electrical Panels (HV Systems, MV Systems & LV Systems)
- Design of Control & Automation Systems
- Design of Solar Energy Systems
- Design of Electric Vehicle Equipment
- Design of Power Quality Systems
- Design of Power protection Systems

9. **Testing, Installation, Commissioning, Maintenance (Repair & Troubleshooting) & Innovation (Redesigning & Upgrades) - Electrical Systems Services**

10. **System Design Engineers / Electrical Schematic Designer (Training, Certifications, Job Description)**